PEIYING YANG

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EDUCATION

Xi'an University of Posts and Telecommunications | Optoelectronic Information Science and Engineering | Bachelor's Degree — Sep 2023 - Jun 2027

Relevant coursework: Physical Optics; Electromagnetic Fields and Waves; Signals and Systems; Circuit Analysis; Semiconductor Physics; Digital Logic; Display Device Driving Circuits.

INTERNSHIP EXPERIENCE

Automotive T-Box Functional Research, Development and Testing

Zhejiang Tsinghua Yangtze River Delta Institute | Partner: Shaanxi Automobile Group

Jun 2026 - Sep 2026

- Participated in research and analysis, development support, and testing and validation of automotive T-Box functions.
- Assisted with functional-module development, program debugging, and functional integration testing according to project requirements.
- Participated in vehicle-terminal functional testing, abnormal-case reproduction, and issue recording.
- Organized, analyzed, and provided feedback on issues identified during testing.
- Coordinated with project personnel from the institute and Shaanxi Automobile Group to complete project development and testing.
- Gained an understanding of the basic workflow of automotive electronics products, from functional research, development and debugging to testing and validation.

PROJECT EXPERIENCE

Shopping Mall Store Management System Development

Project Role: Project Lead

Project Time: Jun 2024 - Jul 2024

Developed a management system for a shopping-mall store, implementing basic product-management and order-management functions.

- Analyzed daily store-operation processes and planned system function modules.
- Designed product-information management and order-management modules.
- Independently completed system coding and functional testing.
- Debugged and modified issues arising during system operation.
- Completed the implementation of basic system functions and defined the operational workflow.

Laser Rangefinder Development Based on STM32F401

Project Role: Laboratory Project Member

Project Time: Sep 2024 - Nov 2024

Participated in the development of a laboratory laser rangefinder, using an STM32F401 development board to develop software for selected functional modules and write the corresponding programs.

- Participated in the design of laser-rangefinder functional modules according to project requirements.
- Completed relevant program development and functional debugging on the STM32F401 development board.
- Participated in the development of peripheral control, data processing, and system-control logic.
- Troubleshot issues, made code modifications, and conducted functional verification during program operation.
- Participated in hardware-software integration testing to help improve system stability.

Digital Quiz Buzzer Hardware and Software Development

Project Role: Project Lead

Project Time: Mar 2025 - Apr 2025

Designed and implemented a digital quiz-buzzer system with answering, timing, display, and result-determination functions.

- Designed system functions and planned the hardware and software solution.
- Used Multisim to simulate and analyze the system circuits.
- Completed digital-logic program development and functional-module design.
- Performed circuit soldering, hardware debugging, and fault diagnosis.
- Tested the answering, timing, display, and reset functions.

Point Light Source Automatic Tracking System Design

Project Role: Project Lead

Project Time: Apr 2025 - May 2025

Designed and developed a hardware-software system capable of automatically detecting and tracking the position of a point light source.

- Designed the overall project solution and planned system functions.
- Completed the design of the photoelectric-sensor detection circuit.
- Developed point-light-source position-detection and automatic-tracking algorithms.
- Completed sensor-signal acquisition, control-program development, and system integration testing.
- Adjusted parameters for tracking accuracy, response speed, and system stability.
- Troubleshot issues and optimized system performance during operation.

150-inch Home Theater Display Solution Design

Project Role: Project Lead

Project Time: May 2025

Designed a complete large-screen and optical-display solution for a 150-inch home-theater scenario.

- Analyzed the usage requirements and system-configuration requirements of the home-theater display scenario.
- Planned the optical path and selected and calibrated display hardware.
- Completed compatibility analysis of projection equipment, display equipment, and related hardware.
- Debugged optical and display performance.
- Produced a complete design solution with practical implementation feasibility.

Intelligent Pickup Navigation System

Project Role: Project Lead

Project Time: Nov 2025 - Dec 2025

Led solution design, control-program development, and function optimization for an intelligent navigation system.

- Designed the overall system architecture and planned functional modules.
- Wrote the navigation-control program to implement the basic system operating logic.
- Optimized the control flow and functional modules based on system operation results.
- Organized project materials, system descriptions, and project-application documents.

Laser Beam Expansion Optical System Simulation Design

Project Role: Project Member

Project Time: Dec 2025 - Jan 2026

Participated in the solution design, optical-path simulation, and experimental-data processing of a laser beam expansion optical system.

- Participated in overall solution design and optical-component selection for the laser beam expansion system.
- Completed optical-path planning analysis and optical-system simulation.
- Organized, analyzed, and visualized simulation results and experimental data.
- Used Origin for data fitting, plotting, and result comparison.
- Participated in writing the complete optical-system design report.

STM32-based Audio Player Design and Development

Project Role: Project Lead

Project Time: May 2026 - Jun 2026

Completed the overall design and development of an STM32-based embedded audio player, including audio playback, graphical interaction, and hardware-system integration testing.

- Designed the overall solution, planned system functions, and built the hardware-software architecture.
- Independently completed STM32 main-control program development, including audio playback, button interaction, and system control.
- Used TouchGFX to design the graphical user interface, including page layout, interaction logic, and functional logic integration.
- Completed SD-card audio-file reading and audio-playback development.
- Participated in PCB production, soldering, hardware debugging, and system integration testing.
- Located, analyzed, and repaired functional abnormalities and hardware faults arising during system operation.

License Plate Detection and Recognition System

Project Role: Project Member

Project Time: Jun 2026 - Jul 2026

Participated in the development of a license-plate detection and recognition system, implementing image acquisition, image preprocessing, license-plate localization, and character recognition.

- Participated in system-development framework design and functional-module division.
- Completed camera configuration and image-acquisition debugging.
- Participated in image preprocessing, including image enhancement, grayscale conversion, and noise processing.
- Participated in license-plate detection and character-recognition algorithm development.
- Debugged and optimized key parameters in the recognition process.
- Completed overall system testing and functional validation.

HONORS

- 2023 Provincial Undergraduate Physics Experiment Competition - Third Prize
- 2024 China International College Students' Innovation Competition - Second Prize at Provincial Level
- 2025 Mathematical Contest in Modeling (MCM/ICM) - H Prize
- 2025 Mathematical Modeling Competition - First Prize

CAMPUS EXPERIENCE

EPI Huawei Laboratory | Member

Sep 2023 - Mar 2025

- Participated in optoelectronic hardware development, embedded programming, and laboratory research projects.
- Participated in laboratory projects and competition tasks, completing system development, hardware debugging, and problem diagnosis.
- Applied optoelectronics knowledge to practical engineering projects and improved hardware-software co-development skills.
- Became familiar with the basic workflow from solution design and program development to system testing.

TECHNICAL SKILLS

Programming and Embedded Systems: C; STM32 development; TouchGFX

Hardware and Circuits: JLCEDA; Multisim; PCB design; soldering and hardware debugging

Optics and Data Processing: optical-system simulation; Origin; optical-data analysis

Imaging and Algorithms: computer vision; image preprocessing; license-plate recognition

Office Software: Microsoft Office

RESEARCH INTERESTS

Optical signal processing; display driving; chromatic-dispersion compensation; low-light CMOS imaging; embedded systems; and digital-system design.